A dense-case theorem for Seymour's second-neighborhood conjecture

Jake Brukhman
jake@coinfund.io

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Abstract

Seymour's Second Neighborhood Conjecture asserts that every finite oriented graph has a vertex with at least as many exact second outneighbors as outneighbors. Established cases include tournaments, proved by Fisher [1], and oriented graphs of minimum outdegree at most six, proved by Kaneko and Locke [2]; a recent preprint treats minimum outdegree seven [3]. For dense incomplete graphs, Fidler and Yuster proved the conjecture when the missing edges form a matching, a star, or a clique [4], and Ghazal extended this direction to generalized stars [5]. We give a short counting proof of the conjecture for every oriented graph of order $n = 2\delta + 2$, where δ is the minimum outdegree, with no prescribed structure on the missing edges. Together with Fisher's tournament theorem, this implies the conjecture for every oriented graph satisfying $n \leq 2\delta + 2$. Combined with the known minimum-outdegree results, this raises the unconditional lower bound on the order of a counterexample from 16 to 17 and, conditional on a recent pre-print from Sadhukhan, et. al., from 18 to 19.

1 Introduction

A finite *directed graph* is a pair $G = (V(G), A(G))$, where $V(G)$ is a finite set of vertices and $A(G)$ is a set of ordered pairs of distinct vertices, called *arcs*. We write

$$u \to v$$

when $(u, v) \in A(G)$. Thus the arrow points from the source of the arc to its target. The directed graph is *oriented* if, for every two distinct vertices u, v, at most one of $u \to v$ and $v \to u$ is present. For a vertex $v \in V(G)$, define its outdegree and indegree by

$$d^+(v) = |\{x \in V(G) : v \to x\}|, \qquad d^-(v) = |\{x \in V(G) : x \to v\}|.$$

The minimum outdegree of G is $\delta = \min_{v \in V(G)} d^+(v)$. For a vertex $v \in V(G)$, define the *exact first outneighborhood* by

$$N_1(v) = \{x \in V(G) : v \to x\},$$

and the *exact second outneighborhood* by

$$N_2(v) = \{x \in V(G) \setminus N_1(v) : y \to x \text{ for some } y \in N_1(v)\}.$$

Note that $N_2(v)$ is disjoint from $N_1(v)$ and also $\{v\}$. Thus $N_2(v)$ consists exactly of the vertices reached from v in two steps but not in one step. When discussing reachability from v to x, we call v the *source vertex* and x the *target vertex*. A vertex v is a *Seymour vertex* if $|N_2(v)| \geq |N_1(v)|$.

Conjecture (Seymour)**.** *Every finite oriented graph has a Seymour vertex.*

Several results bear directly on dense or extremal cases. Fidler and Yuster [4] proved the conjecture for several prescribed missing-edge structures, and Ghazal [5] generalized that line of work. Espuny Díaz, Girão, Granet, and Kronenberg [6] obtained reductions involving minimum outdegree, and Guo, Kang, and Zwaneveld [7] studied Seymour-tight orientations and counterexamples close to regular tournaments.

Let G be an oriented graph on n vertices with minimum outdegree δ. Counting arcs by their sources gives

$$n\delta \leq |A(G)| \leq \binom{n}{2},$$

so $n \geq 2\delta + 1$. Equality forces every pair of vertices to be adjacent and every outdegree to equal δ, so the graph is a regular tournament which we know satisfies the conjecture by Fisher's theorem.

Theorem 1.1 (Fixed-target theorem)**.** *Let G be an oriented graph on n vertices with minimum outdegree δ. If*

$$n = 2\delta + 2,$$

then G has a Seymour vertex.

At $n = 2\delta + 2$, up to $n/2$ edges may be missing and no structure is imposed on them, in contrast to [4, 5]; conversely, those results impose no lower bound on the minimum outdegree, so neither result contains the other. Corollary 1.2 shows that any counterexample to the conjecture must satisfy $n \geq 2\delta + 3$: counterexamples, if any exist, can be made close to regular tournaments in the sense of [7], but they cannot occur at the two densest orders an oriented graph of minimum outdegree δ can have.

Corollary 1.2 (Dense-case corollary)**.** *Let G be an oriented graph on n vertices with minimum outdegree δ. If $n \leq 2\delta + 2$, then G has a Seymour vertex.*

Proof. Immediately from Fisher and Theorem 1.1. $\square$

Corollary 1.3 (Minimum counterexample order)**.** *Every counterexample to Seymour's conjecture has at least* 17 *vertices. Assuming the minimum-outdegree-seven result of Sadhukhan, Sandeep, and Sen [3], every counterexample has at least* 19 *vertices.*

Proof. By Kaneko and Locke's theorem [2], a counterexample has minimum outdegree $\delta \geq 7$. Corollary 1.2 then gives $n \geq 2\delta + 3 \geq 17$. Under the result of Sadhukhan, Sandeep, and Sen, the case $\delta = 7$ is also excluded, so $\delta \geq 8$ and hence $n \geq 19$. $\square$

2 The fixed-target argument

For a source vertex v, define

$$U(v) = V(G) \setminus \Big(\{v\} \cup N_1(v) \cup N_2(v)\Big).$$

Thus $U(v)$ is the set of target vertices that the source v cannot reach in at most two steps. Now fix a target vertex x. Define the *trap* of x by

$$T(x) = \{y \in V(G) \setminus \{x\} : y \nrightarrow x\}.$$

These are exactly the vertices that do not send an arc to x. Because G is oriented, every outneighbor of x belongs to $T(x)$. Hence

$$|T(x)| \geq d^+(x) \geq \delta.$$

Define the *target slack*

$$q(x) = |T(x)| - \delta \geq 0.$$

Here both $|T(x)|$ and δ count vertices: δ is the minimum number of outneighbors at any vertex. Thus $q(x)$ is the number of vertices by which the trap exceeds the size of a minimum outneighborhood. Define

$$P(x) = \{v \in V(G) : x \in U(v)\}.$$

Thus $P(x)$ is the set of source vertices from which the fixed target x cannot be reached in at most two steps. If $v \in P(x)$, then $v \nrightarrow x$. Moreover, no vertex of $N_1(v)$ points to x, since otherwise there would be a two-step path from v to x. Therefore

$$\{v\} \cup N_1(v) \subseteq T(x). \tag{1}$$

This is the fixed-target observation: every source v that cannot reach x in 2 steps, together with all of its outneighbors, is confined to the same trap $T(x)$.

Lemma 2.1 (Fixed-target capacity)**.** *Let x be a target vertex. If $P(x) \neq \varnothing$, then*

$$q(x) \geq 1 \qquad \text{and} \qquad |P(x)| \leq 2q(x) - 1.$$

In particular, if $q(x) = 0$, then $P(x) = \varnothing$.

Proof. Suppose $P(x) \neq \varnothing$, and set $p := |P(x)| \geq 1$. By (1), $P(x) \subseteq T(x)$, and every arc whose source lies in $P(x)$ has its target in $T(x)$. There are at least δp such arcs. At most $\binom{p}{2}$ have both endpoints in $P(x)$, because G is oriented, and at most $p\Big(|T(x)| - p\Big)$ go from $P(x)$ to the rest of the trap. Consequently,

$$\delta p \leq \binom{p}{2} + p\Big(\delta + q(x) - p\Big).$$

Dividing by p and rearranging gives $p \leq 2q(x) - 1$. Since $p \geq 1$, this forces $q(x) \geq 1$. The final assertion follows. $\square$

The point of the lemma is that target slack is the only room available for packing many source vertices that all miss the same target. Suppose for contradiction that G has no Seymour vertex. Then $\delta > 0$, since a vertex of outdegree zero is automatically a Seymour vertex. Let

$$L = \{v \in V(G) : d^+(v) = \delta\}, \qquad \ell = |L|.$$

Some vertex attains the minimum outdegree, so $\ell \geq 1$. Every $v \in L$ is not a Seymour vertex, so $|N_2(v)| \leq \delta - 1$. Since $n = 2\delta + 2$, it follows that

$$|U(v)| = n - 1 - \delta - |N_2(v)| \geq 2. \tag{2}$$

Thus each minimum-outdegree source contributes at least two unreachable targets. Let

$$I = \Big|\{(v, x) \in V(G)^2 : x \in U(v)\}\Big|.$$

Counting these ordered source–target pairs first by their source and then by their target gives

$$I = \sum_{v \in V(G)} |U(v)| = \sum_{x \in V(G)} |P(x)| \tag{3}$$

since $x \in U(v)$ and $v \in P(x)$ describe the same ordered pair (v, x). From (2),

$$I \geq 2\ell > 0. \tag{4}$$

Put

$$Q = \sum_{x \in V(G)} q(x).$$

Since $I > 0$, at least one set $P(x)$ is nonempty; Lemma 2.1 then implies that at least one $q(x)$ is positive. Applying the lemma in (3) gives

$$I \leq \sum_{q(x)>0} \Big(2q(x) - 1\Big) < 2\sum_{x} q(x) = 2Q. \tag{5}$$

So the total demand is strictly less than twice the total target slack. It remains to compare Q with ℓ. Since $|T(x)| = n - 1 - d^-(x)$,

$$q(x) = n - 1 - \delta - d^-(x).$$

Total indegree equals total outdegree, and every vertex outside L has outdegree at least $\delta+1$. Therefore

$$\begin{aligned} Q &= n(n - 1 - \delta) - \sum_{x} d^+(x) \\ &\leq n(n - 1 - \delta) - \Big(\ell\delta + (n - \ell)(\delta + 1)\Big) \\ &= \ell, \end{aligned} \tag{6}$$

where the last equality uses $n = 2\delta + 2$. This degree ledger bounds the total room in all target traps using only the number of minimum-outdegree vertices. Now (4)–(6) give

$$I < 2Q \leq 2\ell \leq I,$$

a contradiction. Hence G has a Seymour vertex.

Remark. The bound $n \leq 2\delta + 2$ is exactly where this argument stops. For $n = 2\delta + 3$, the same ledger calculation gives only $Q \leq \ell + n$, while the demand calculation gives $I \geq 3\ell$; since $\ell \leq n$, the comparison $I < 2Q \leq 2\ell + 2n$ no longer yields a contradiction. Extending the method past $n = 2\delta + 2$ therefore requires new ideas.

AI contribution and provenance

Attribution for this work is as follows. The author initiated and directed the investigation, selected its falsification-first and structural strategy, curated intermediate results, chose the theorem for publication, and edited the final statement and exposition. OpenAI language models—principally a Codex GPT-5-family multi-agent run and GPT-5.6 Sol—conducted the detailed mathematical exploration, implemented counterexample searches and verification tools, discovered the fixed-target capacity argument and its double-counting proof, and drafted the manuscript. Claude Fable 5 from Anthropic was used for an adversarial audit of an intermediate draft. The author accepts sole responsibility for the final manuscript and its claims.

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